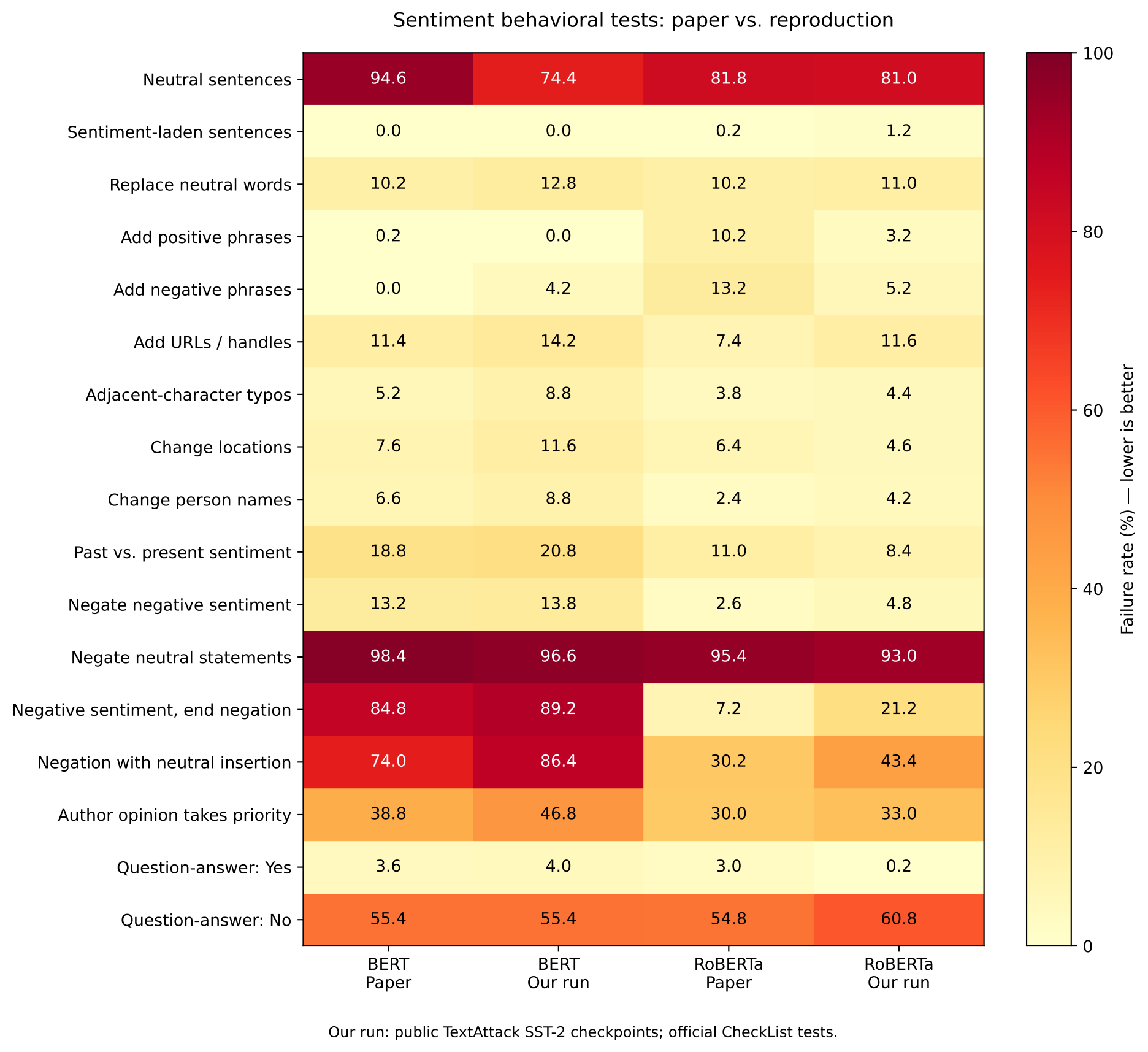
Sentiment behavioral tests: paper vs. reproduction
Neutral sentences
Sentiment-laden sentences
Replace neutral words
Add positive phrases
Add negative phrases
Add URLs / handles
Adjacent-character typos
Change locations
Change person names
Past vs. present sentiment
Negate negative sentiment
Negate neutral statements
Negative sentiment, end negation
Negation with neutral insertion
Author opinion takes priority
Question-answer: Yes
Question-answer: No
94.6 74.4 81.8 81.0
0.0 0.0 0.2 1.2
10.2 12.8 10.2 11.0
0.2 0.0 10.2 3.2
0.0 4.2 13.2 5.2
11.4 14.2 7.4 11.6
5.2 8.8 3.8 4.4
7.6 11.6 6.4 4.6
6.6 8.8 2.4 4.2
18.8 20.8 11.0 8.4
13.2 13.8 2.6 4.8
98.4 96.6 95.4 93.0
84.8 89.2 7.2 21.2
74.0 86.4 30.2 43.4
38.8 46.8 30.0 33.0
3.6 4.0 3.0 0.2
55.4 55.4 54.8 60.8
BERT Paper
BERT Our run
RoBERTa Paper
RoBERTa Our run
Failure rate (%) — lower is better
0
20
40
60
80
100
Our run: public TextAttack SST-2 checkpoints; official CheckList tests.